

7. Integration Techniques

7.1. Integration by Parts

7.1a Learning Objectives

- I understand how integration by parts compares to the product rule.
- I understand when and how to use integration by parts.
- I can choose an appropriate u and v .
- I can use integration by parts to reduce integrals.

Recall that u -substitution corresponded to “undoing” the chain rule (when taking derivatives). What corresponds to the product rule?

Product Rule:

$$\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + f'(x)g(x)$$

Integrating both sides, and rearranging slightly, we end up with the following

$$\begin{aligned}\frac{d}{dx}[f(x)g(x)] &= f(x)g'(x) + f'(x)g(x) \\ f(x)g(x) + C &= \int f(x)g'(x) dx + \int f'(x)g(x) dx \\ f(x)g(x) - \int f'(x)g(x) dx &= \int f(x)g'(x) dx\end{aligned}$$

Relabeling the last line with a few substitutions $u = f(x) \Rightarrow du = f'(x) dx$ and $v = g(x) \Rightarrow dv = g'(x) dx$ we end up with

Theorem 7.1.1 Integration by Parts

$$\int u dv = uv - \int v du$$

Example 7.1.1

Find the antiderivative:

$$\int x \sin(x) dx$$

Solution

We don't know a best method for picking u and dv yet, so let's try something and see what happens. Let

$$\begin{aligned}
 u &= \sin(x), & dv &= x \, dx, \\
 du &= \cos(x) \, dx, & v &= \int x \, dx \\
 & & &= \frac{x^2}{2}
 \end{aligned}$$

Then

$$\int x \sin(x) \, dx = \sin(x) \frac{x^2}{2} - \underbrace{\int \frac{x^2}{2} \cos(x) \, dx}_{\text{this integral looks worse}}$$

Try the other way around now

$$\begin{aligned}
 u &= x, & dv &= \sin(x), \\
 du &= dx, & v &= \int \sin(x) \\
 & & &= -\cos(x)
 \end{aligned}$$

This gives us

$$\begin{aligned}
 \int x \sin(x) \, dx &= -x \cos(x) - \underbrace{\int -\cos(x) \, dx}_{\text{which we know}} \\
 &= -x \cos(x) + \sin(x) + C
 \end{aligned}$$

Exercise 7.1.1

Check the above answer by taking the derivative. Did you get what we start with?

7.1.1. How to we choose which function to use for u ?

i Note

Recall that u is a function of which we can easily derive, but not necessarily integrate. We tend to pick things that simplify a lot when we take the derivative of them: logarithms have $\frac{d}{dx} \log(x, b) = \frac{1}{\ln(b)x}$, inverse trig functions have become almost algebraic like $\frac{d}{dx} \arcsin(x) = \frac{1}{\sqrt{1-x^2}}$, and polynomials which decrease in number of term and power the more you derive them. Alternatively, things that don't change very much like trig functions and exponential functions might be avoided for u .

Here is a way to discern potential functions for u , keeping in mind that you often have to try a few things as you get used to the process.

1. **I**: Inverse trig functions
2. **L**: logarithmic functions
3. **A**: algebraic functions (polynomials, rational functions)
4. **T**: trig functions
5. **E**: exponential functions

7.1.2. Indefinite Integrals

Example 7.1.2

Find

$$\int 2xe^x \, dx.$$

Solution

If we consider taking the exponential term as our choice for u , then when we take the derivative of this term, it stays exactly the same. On the other hand, if we let $u = 2x$ then when we take the derivative we get something simpler. This leads us to want to try $u = 2x$ first. (But we might have to try something else later)

Then,

$$\begin{aligned} u &= 2x, & dv &= e^x \, dx, \\ du &= 2 \, dx, & v &= \int e^x \, dx = e^x \end{aligned}$$

gives us

$$\begin{aligned} \int 2xe^x \, dx &= 2xe^x - \int 2e^x \, dx \\ &= 2xe^x - 2e^x + C \end{aligned}$$

Example 7.1.3

Find

$$\int e^{\sqrt{x}} \, dx$$

Solution

This problem is a little tricky, it needs a substitution first, and then it can be integrated by parts. Let $w = \sqrt{x}$ then to get dx we can solve for x : $w^2 = x$ so $2w \, dw = dx$. Then the integral becomes

$$\int e^{\sqrt{x}} \, dx = \int 2we^w \, dw$$

Which we solved in the previous problem. So,

$$\int e^{\sqrt{x}} \, dx = 2\sqrt{x}e^{\sqrt{x}} - 2e^{\sqrt{x}} + C$$

Exercise 7.1.2

Find

$$\int x^2 e^x \, dx.$$

Tip

Hint: Try something.

7.1a Section Summary:

- We learned how to “undo” the product rule, via integration by parts.
- We learned to use substitution and integration by parts in the same problem.
- We learned that sometimes you have to do integration by parts multiple times to get a final answer.